

RESEARCH SPECIFICATION • VERSION 0.3

Gas storage as a curve-dependent real option.

A mathematical foundation for valuing physical inventory flexibility, learning operating policies, and separating forward-curve hedges from basis and operational risk.

$$V_t(s) = \sup_{\pi \in \Pi} E_t^Q \left[\sum_{u=t}^{T-1} D_{t,u} C_u(s_u, \pi_u(s_u)) + D_{t,T} G(s_T) \right] \quad (0)$$

🕒 Monthly toy model today

✓ Simulated sensitivity plots added

[Read the specification](#) ↓

01 • VALUATION PROBLEM

Contract, measure, and objective

The storage is a finite-horizon stochastic control problem. Physical inventory is controlled through injection, withdrawal, or hold decisions; market risk enters through the forward curve and local basis.

Let decision dates be $t_0 < \dots < t_N = T$. At each date, the operator observes market and facility states s_t , chooses a feasible control q_t , receives cashflow C_t , and carries inventory into the next date. The valuation measure must be stated explicitly:

Risk-neutral value

Use Q with curves and tradable-factor dynamics calibrated to option and futures markets. This is the primary fair-value specification.

Real-world strategy

Use P for expected operating P&L, hedge backtests, liquidity costs, and risk limits. A statistical forecast is not automatically a fair-value model.

MODELING JUDGMENT

Report fair value and strategy value separately. Mixing historical drift with risk-neutral discounting can create apparent value that is really a directional forecast.

02 • SIMULATED MODEL LAB

A runnable toy valuation workflow

The current code generates simulated state paths, fits an LSMC policy, evaluates it out of sample, and publishes diagnostics that explain which assumptions move value. These are research checks, not calibrated market marks.

BASE LSMC VALUE

178.393

standard error 2.253

LARGEST UPSIDE TEST

Higher prompt vol

value 192.145

LARGEST DOWNSIDE TEST

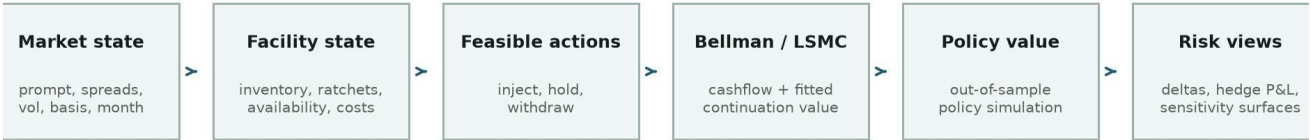
Tighter terminal target

value 158.681

VOL / MEAN-REVERSION RANGE

167.108-175.538

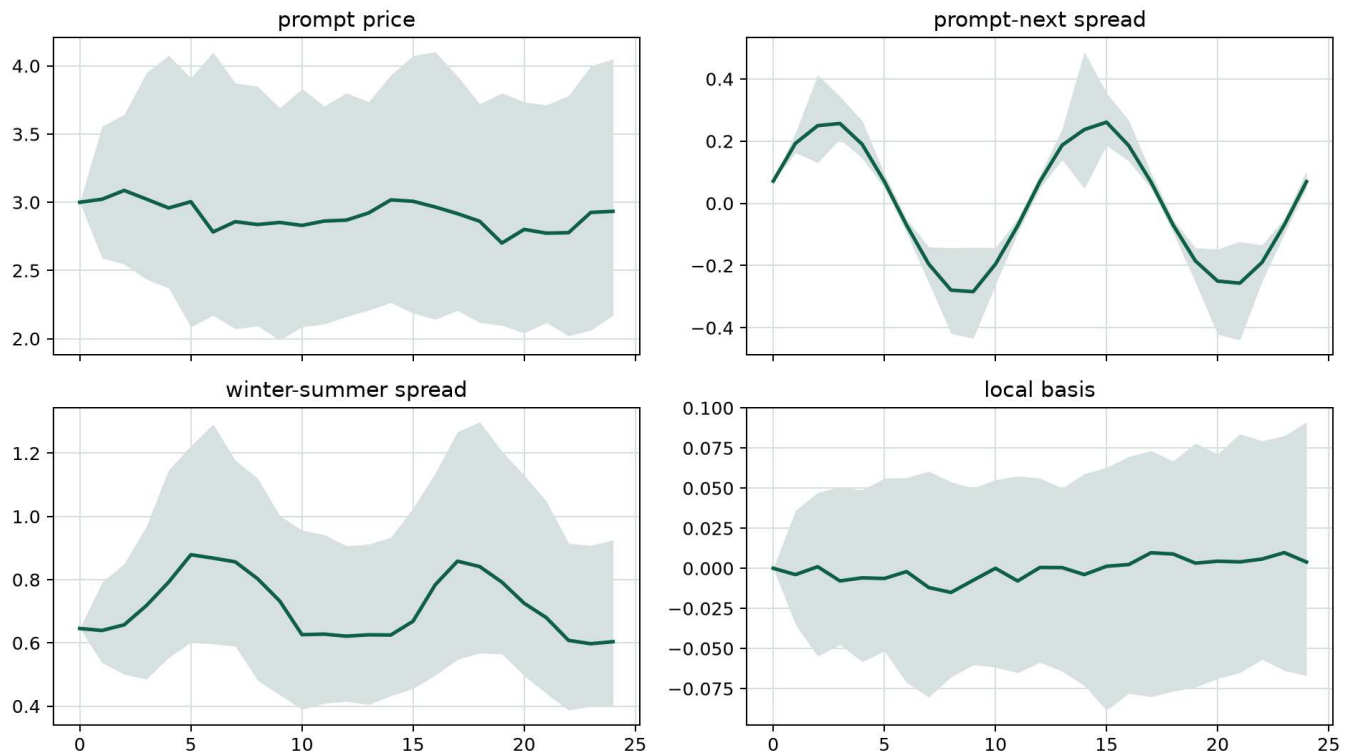
surface min to max



One environment, common random numbers, frozen policy evaluation

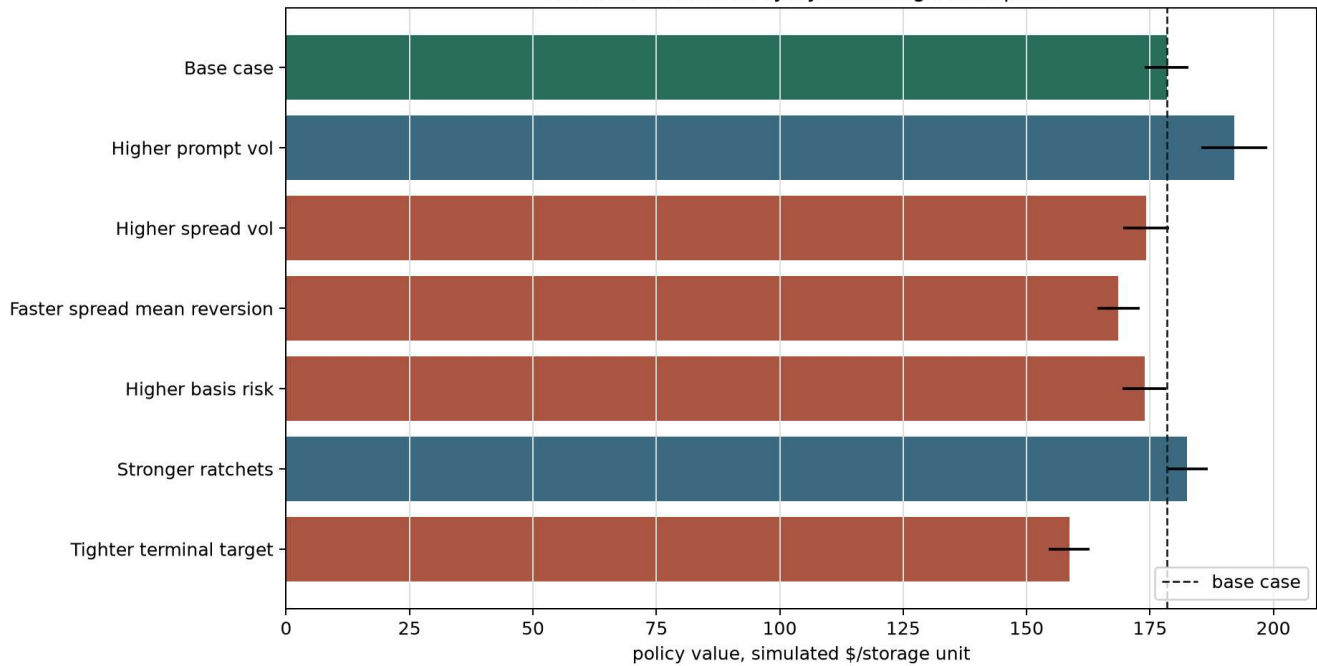
The website now follows the same modeling pipeline as the Python package: one simulated environment, common random numbers, fitted policy evaluation, and then risk diagnostics.

Simulated state variables used by the policy

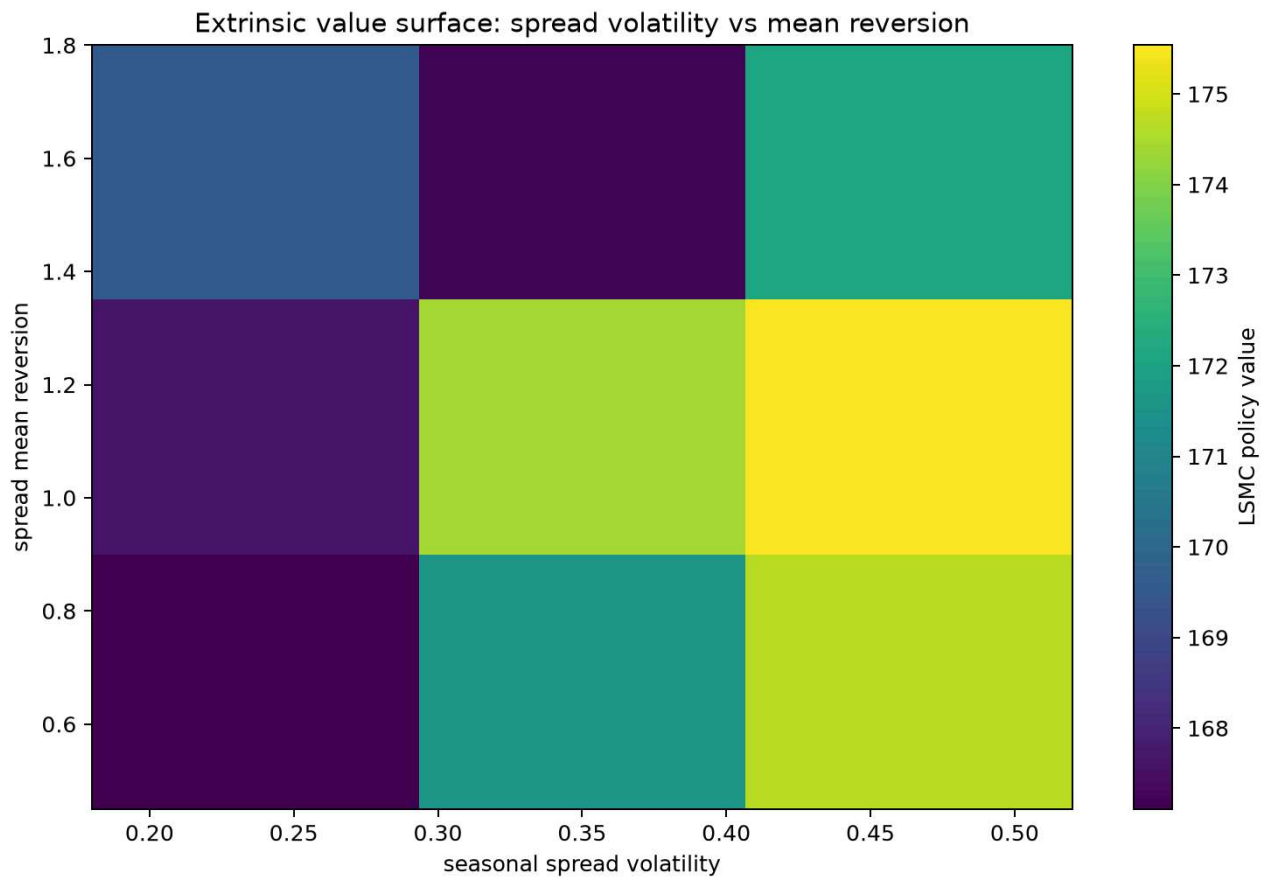


The policy observes prompt price, short spread, seasonal spread, basis, volatility proxies, calendar state, and normalized inventory.

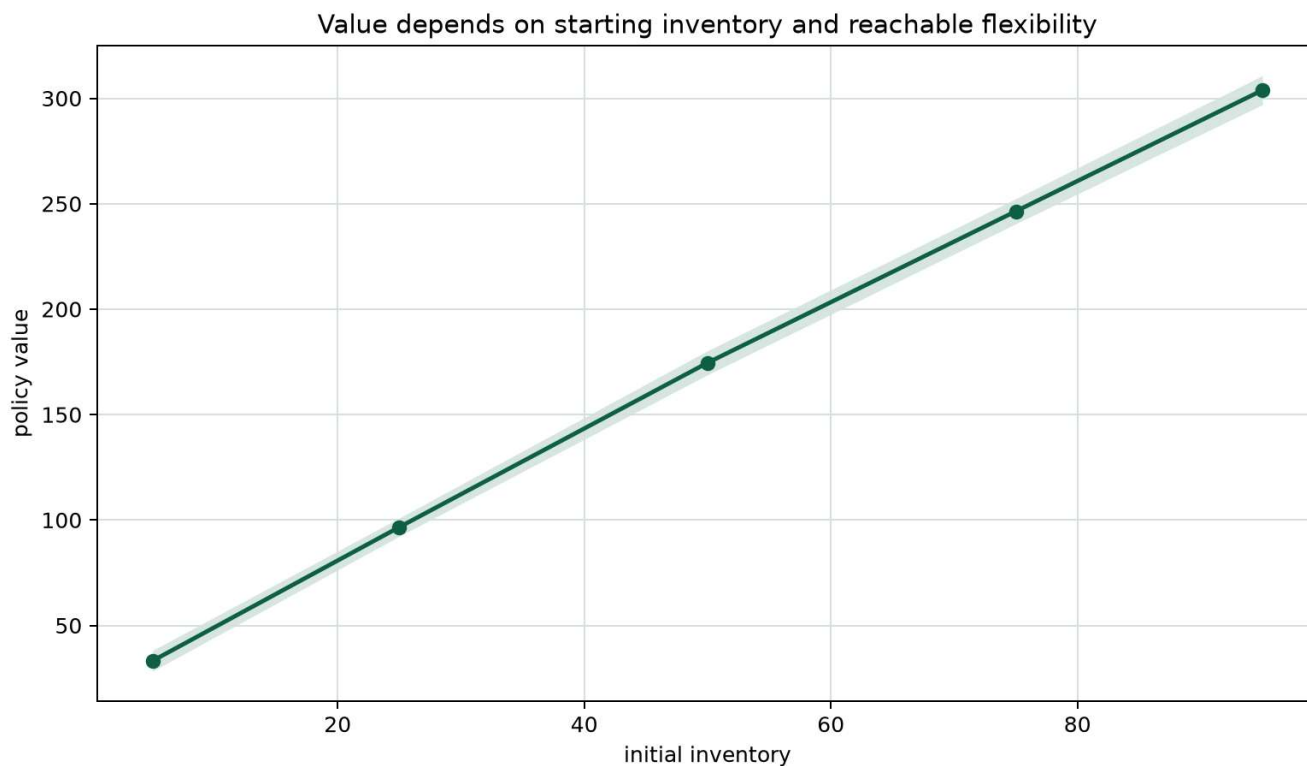
LSMC value sensitivity by modeling assumption



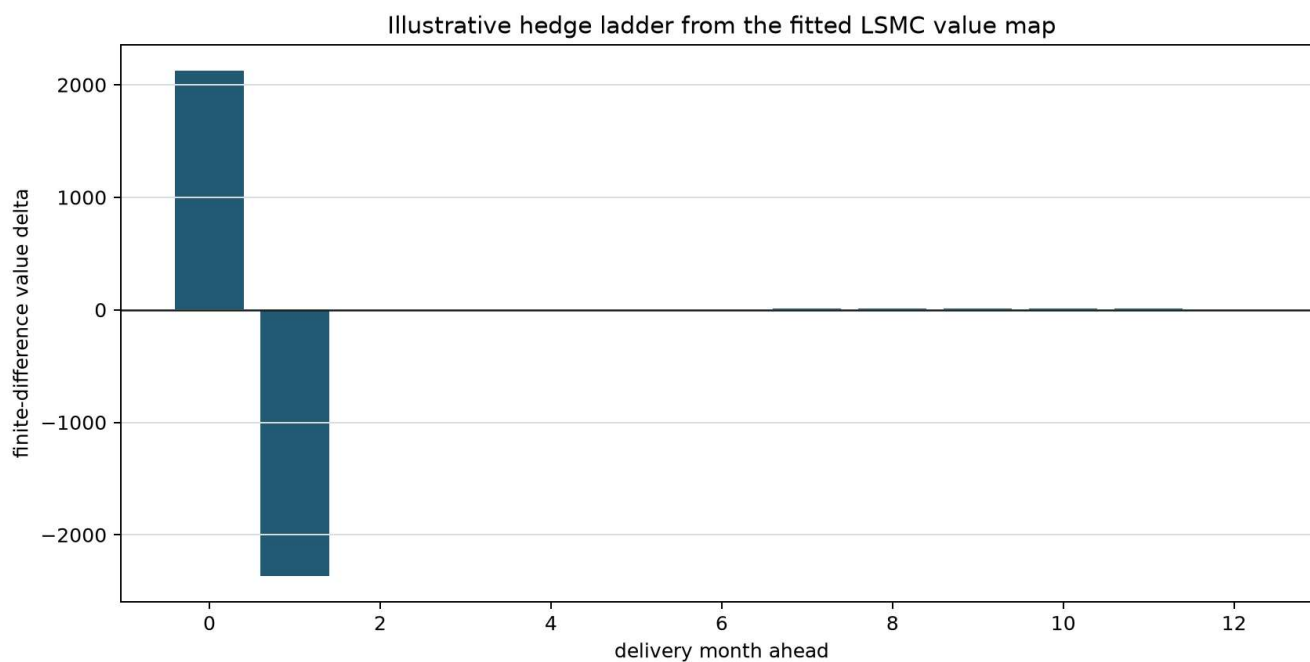
The first sensitivity pass shows that terminal constraints, mean reversion, basis risk, and ratchets can move value as much as volatility bumps.



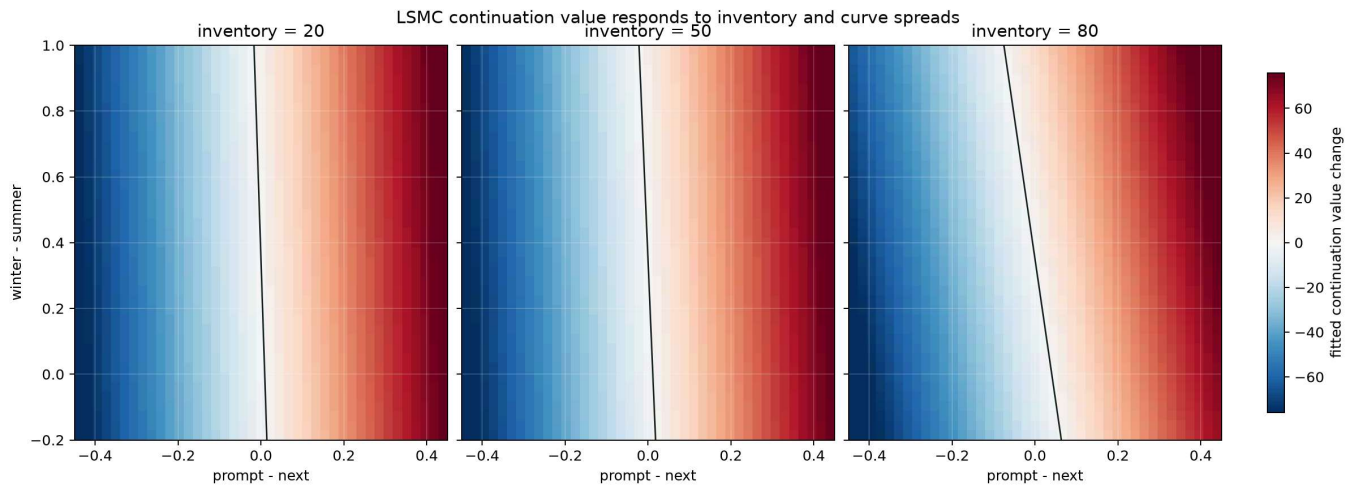
This surface is the planned template for extrinsic-value studies: vary spread volatility and mean reversion jointly rather than reporting a single volatility shock.



Initial inventory changes the reachable set of future states. The same forward curve can have different value depending on whether injection or withdrawal flexibility is scarce.



The hedge ladder is finite-difference based on the fitted value map. The production version should bump calibrated factors and map them to tradable monthly instruments.



The continuation-value surface makes the inventory-spread interaction visible. This is a regression diagnostic: a policy chart that collapses to one action is a warning, not an insight.

FINDING 1

Volatility is not a scalar value driver. Prompt volatility, seasonal spread volatility, and correlation with basis should be separated.

FINDING 2

Ratchets and terminal rules control whether flexibility can actually be monetized. Higher volatility without reachable inventory states may not add much value.

FINDING 3

Basis risk must be carried into hedge diagnostics. Hub deltas can look clean while local physical P&L remains exposed.

CURRENT CODE BOUNDARY

The simulation uses fast path counts and smooth toy ratchets so it can run on a laptop. The final version should add calibrated market curves, daily facility rules, cross-fit LSMC, richer RL training, and hedge backtests before any chart is treated as a valuation result.

02 • STATE

Inventory meets the observable curve

The Markov state must be rich enough to explain future cashflows and feasible controls, but compact enough for stable conditional-value estimation.

$$s_t = (\bar{I}_t, F_t^{(0)}, D_t^{pn}, D_t^{ws}, B_t, \sigma_t^p, \sigma_t^s, m_t, A_t)$$

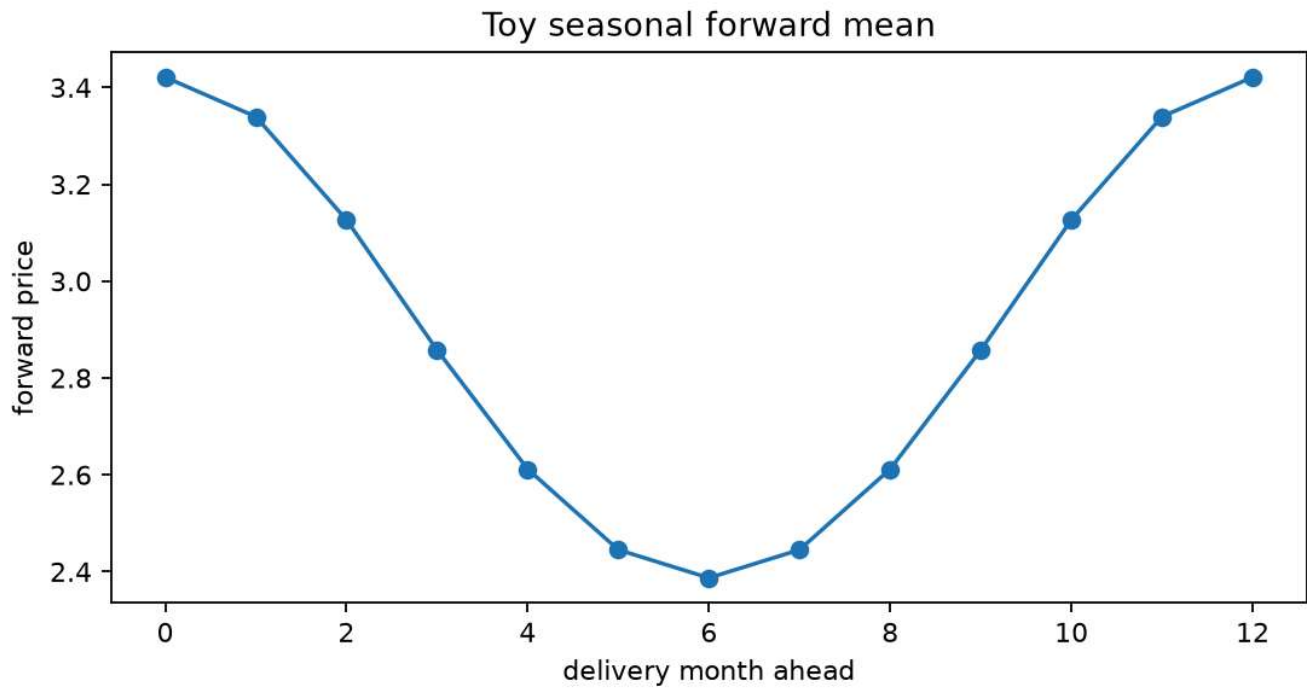
(1)

Recommended state used by both the LSMC and RL formulations.

$\bar{I}_t$	Normalized inventory, $I_t/I_{\max}$
$F_t^{(0)}$	Prompt hub forward
D_t^{pn}	Prompt minus next month
D_t^{ws}	Winter average minus summer average
B_t	Local physical basis to the hedge hub
σ_t^p, σ_t^s	Prompt and spread volatility states
m_t	Month or season indicators
A_t	Availability, outage, and constraint state

$$D_t^{pn} = F(t, T_0) - F(t, T_1), \quad D_t^{ws} = \frac{1}{|W|} \sum_{j \in W} F(t, T_j) - \frac{1}{|S|} \sum_{j \in S} F(t, T_j)$$

(2)



Illustrative simulated curve. The shape supplies value signals; it is not a market calibration or a valuation result.

03 • MARKET DYNAMICS

A multifactor forward-curve model

A practical valuation model should move prompt, long-term level, seasonal spreads, volatility, and local basis independently but with calibrated correlation.

Recommended risk-neutral specification

For delivery date T_j , use a multifactor lognormal forward model. Because the modeled instruments are forwards, the risk-neutral drift is zero before convexity and settlement adjustments.

$$\frac{dF(t, T_j)}{F(t, T_j)} = \sigma_L(T_j) dW_t^L + \sigma_P e^{-\kappa_P(T_j-t)} dW_t^P + \sigma_S g_S(T_j) dW_t^S \quad (3)$$

The long factor moves the whole curve, the decaying prompt factor creates Samuelson-style front volatility and mean reversion, and the seasonal loading $g_S(T)$ moves winter against summer. Under correlated shocks with instantaneous correlation matrix R_t , the exact one-step simulation is:

$$\log \frac{F(t + \Delta t, T_j)}{F(t, T_j)} = -\frac{1}{2} \Sigma_j(t) \Delta t + \sigma_j(t)^\top L_t \sqrt{\Delta t} \varepsilon_{t+1}, \quad L_t L_t^\top = R_t \quad (4)$$

$$\Sigma_j(t) = \sigma_j(t)^\top R_t \sigma_j(t), \quad \varepsilon_{t+1} \sim N(0, I) \quad (5)$$

Stochastic volatility and spread variance

$$dz_t = \kappa_v(\theta_v - z_t)dt + v_v \sqrt{z_t} dW_t^v, \quad \sigma_j(t) = \sqrt{z_t} \sigma_j^0(t) \quad (6)$$

Storage responds to relative prices, so the variance of a spread matters more directly than the variance of either leg in isolation:

$$\text{Var}_t(\Delta F_i - \Delta F_j) = \sigma_i^2 + \sigma_j^2 - 2\rho_{ij}\sigma_i\sigma_j \quad (7)$$

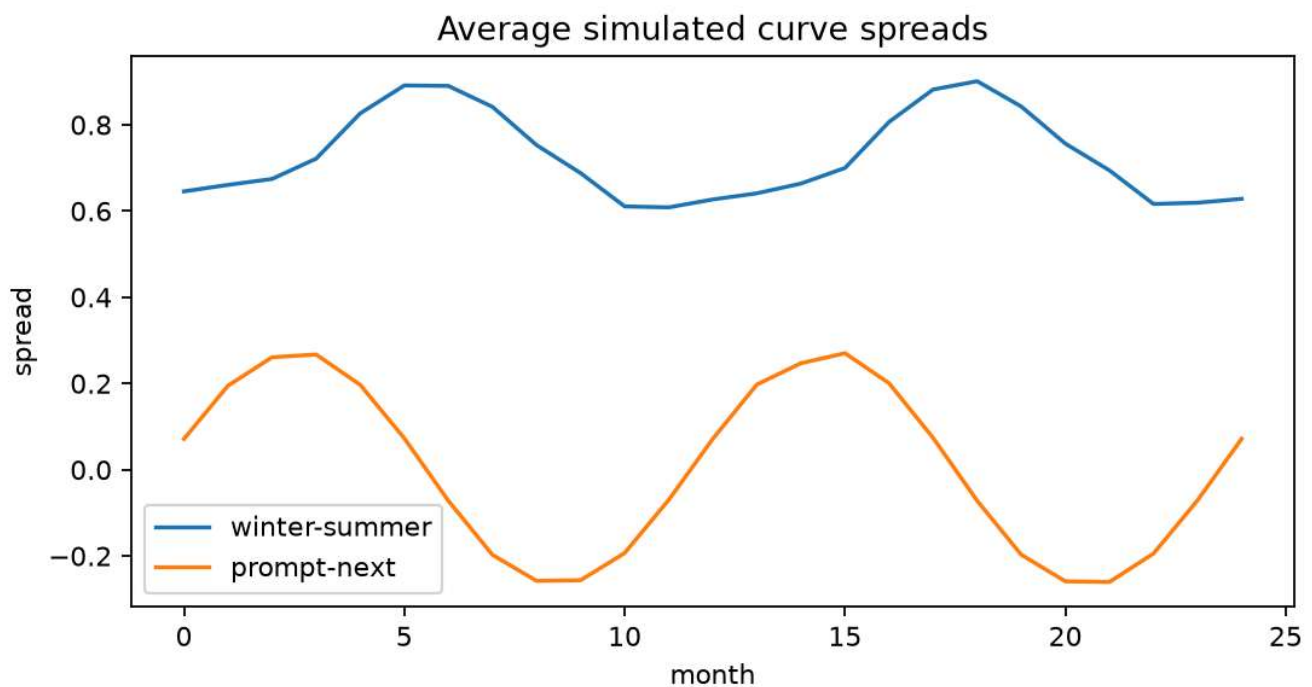
IMPLICATION

Raising prompt volatility does not guarantee a higher storage value. If both delivery legs move together, actionable spread volatility can remain small; stronger mean reversion and ratchet access can matter more.

Local basis

$$P_t^{loc} = F_t^{(0)} + B_t, \quad dB_t = \kappa_B(\theta_B - B_t)dt + \sigma_B dW_t^B + J_t^B dN_t \quad (8)$$

A mean-reverting basis factor captures ordinary location risk; an optional jump term represents congestion or operational dislocations. Basis may be correlated with hub prices, weather, and storage availability.



Illustrative state paths. Final calibration will target the covariance and mean-reversion of tradable monthly spreads and the unhedgeable local basis.

04 • FACILITY

Inventory dynamics, losses, and ratchets

Physical flexibility is state dependent. The same price signal has different value when the cavern is empty, full, slow, unavailable, or close to a ratchet breakpoint.

$$I_{t+1} = (1 - \ell_t)I_t + \eta_{in}q_t^+ - \frac{q_t^-}{\eta_{out}}, \quad 0 \leq I_{t+1} \leq I_{\max} \quad (9)$$

Here $q_t^+, q_t^- \geq 0$ are scheduled injection and withdrawal, η_{in}, η_{out} are efficiencies, and ℓ_t is inventory loss or fuel usage. Simultaneous injection and withdrawal is excluded.

$$0 \leq q_t^+ \leq A_t r_{in}(I_t), \quad 0 \leq q_t^- \leq A_t r_{out}(I_t), \quad q_t^+ q_t^- = 0 \quad (10)$$

Production ratchets should be contract tables, interpolated piecewise linearly rather than globally smoothed. With knots (I_k, r_k) :

$$r(I) = r_k + \frac{r_{k+1} - r_k}{I_{k+1} - I_k}(I - I_k), \quad I \in [I_k, I_{k+1}] \quad (11)$$

Injection optionality

Largest at low inventory, but only if future withdrawal capacity remains sufficient to monetize the gas.

Withdrawal optionality

Largest at high inventory, but constrained by deliverability, minimum inventory, and terminal obligations.

Cashflow and terminal conditions

Cashflows must settle against the physical location while hedges settle against tradable hubs. Terminal treatment should reflect the actual contract, not serve as a numerical convenience.

$$C_t = -q_t^+ (P_t^{loc} + c_{in,t}) + q_t^- (P_t^{loc} - c_{out,t}) - c_{fix,t} - c_{switch} \mathbf{1}_{a_t \neq a_{t-1}}$$

(12)

Three common terminal specifications cover most research cases:

Liquidation

$$G(I_T) = \alpha P_T^{loc} I_T$$

Target penalty

$$G(I_T) = \alpha P_T^{loc} I_T - \lambda_T |I_T - I^*|$$

Hard cycle

$$I_T = I^*$$

VALIDATION CHECK

Re-run the valuation under each economically plausible terminal rule. A large change is not numerical noise; it indicates material residual inventory option value or an underspecified commercial obligation.

The Bellman recursion

Every method in this project solves the same constrained control problem. LSMC approximates continuation values; RL approximates a value or policy map.

$$V_t(s_t) = \max_{a \in A(s_t)} \{C_t(s_t, a) + D_{t,t+1} E_t^Q[V_{t+1}(s_{t+1}) | s_t, a]\} \quad (13)$$

$$V_T(s_T) = G(I_T, P_T^{loc}), \quad D_{t,u} = \exp\left(-\int_t^u r_v dv\right) \quad (14)$$

A useful implementation works with the post-decision states s_t^a : inventory is updated by the chosen flow while the next market shock has not yet arrived. This makes the continuation target cleaner and exposes action-specific feasibility.

07 • REGRESSION MONTE CARLO

LSMC with curve and inventory interactions

The core regression should encode the economics of storage without relying on an uncontrolled polynomial expansion of every forward tenor.

$$\widehat{C}_t(s_t^a) = \phi(s_t^a)^\top \widehat{\beta}_t, \quad \widehat{\beta}_t = \arg \min_{\beta} \sum_{n \in T} (Y_{t+1}^{(n)} - \phi(s_t^{a,(n)})^\top \beta)^2 + \lambda \|\beta\|_2^2 \quad (15)$$

$$\phi = [1, \bar{I}, \bar{I}^2, \log F^{(0)}, D^{pn}, D^{ws}, B, \sigma^p, \sigma^s, \bar{I}D^{pn}, \bar{I}D^{ws}, \bar{I}B, \mathbf{1}_{month}, \mathbf{1}_{season}] \quad (16)$$

Features are standardized within each time step; one month dummy is omitted.

The inventory-spread interactions are structural, not decorative. A positive winter-summer spread can support injection when the cavern is empty, but can have little incremental value at full inventory because injection capacity has vanished.

$$\widehat{Q}_t(s, a) = C_t(s, a) + D_{t,t+1} \widehat{C}_t(s_t^a), \quad \widehat{\pi}_t(s) = \arg \max_{a \in A(s)} \widehat{Q}_t(s, a) \quad (17)$$

Bias controls

- 01 **Cross-fit.** Estimate continuation on one path fold and exercise on another to reduce in-sample look-ahead bias.
- 02 **Use common paths.** Compare policies and volatility scenarios with common random numbers.
- 03 **Evaluate forward.** Train backward, then value the frozen policy on fresh paths with confidence intervals.
- 04 **Stress the basis.** Inspect condition numbers, regularization, fold dispersion, and action-boundary stability.

08 • REINFORCEMENT LEARNING

The same problem as a constrained MDP

RL becomes useful when the action space, constraint state, or objective is too nonlinear for a compact continuation regression. It still requires the same market model and an out-of-sample valuation protocol.

$$\mathbf{M} = (\mathbf{S}, \mathbf{A}, P, r, \gamma), \quad r_t = D_{0,t} C_t, \quad r_T = D_{0,T} G(I_T, P_T^{loc}) \quad (18)$$

For a continuous actor output $u_t \in [-1, 1]$, impose physical feasibility through a deterministic action map:

$$q_t(u_t, I_t) = \begin{cases} u_t r_{in}(I_t), & u_t \geq 0, \\ u_t r_{out}(I_t), & u_t < 0, \end{cases} \quad \text{then clip to inventory bounds} \quad (19)$$

$$J(\theta) = \mathbb{E}[\sum_{t=0}^T \gamma^t r_t], \quad \pi_{\theta}(s) = \arg \max_a Q_{\theta}(s, a) \quad (20)$$

METHOD CHOICE

DQN is a transparent baseline for inject / hold / withdraw. PPO or SAC is more natural once partial rates, continuous nominations, and nonlinear fuel or switching costs are included. Algorithm complexity is not a substitute for a correctly calibrated environment.

09 • OPTIONALITY

Intrinsic and extrinsic value

Extrinsic value is the incremental value of adapting future actions to evolving prices, evaluated with identical physical constraints and terminal treatment.

$$V_0^{intr} = \max_{\{q_t\}} \sum_{t=0}^{T-1} D_{0,t} C_t(q_t; F(0, T_t)) + D_{0,T} G(I_T) \quad (21)$$

$$V_0^{ext} = V_0^{stoch} - V_0^{intr} \quad (22)$$

Prompt volatility

Short-horizon timing

Value vs front-factor vol bump

Spread volatility

Seasonal cycling

Value vs winter-summer covariance

Mean reversion

Repeatable dislocations

Two-way sensitivity; not assumed monotone

Inventory flexibility

Reachable future states

Value by initial inventory and ratchet

Basis risk

Physical / hedge mismatch

Unhedged P&L and basis stress

Outages

Lost exercise opportunities

Availability scenario loss

10 • HEDGE AND RISK

From value to a tradable hedge ladder

Storage is hedged with monthly or seasonal forwards, while basis and physical availability remain only partially hedgeable.

Forward deltas

$$\Delta_{t,j} = \frac{\partial V_t}{\partial F(t, T_j)} \approx \frac{V_t(F + \varepsilon e_j) - V_t(F - \varepsilon e_j)}{2\varepsilon} \quad (23)$$

Bump-and-revalue should preserve the calibrated factor structure and use common random numbers. A raw independent bump to one tenor can create a curve shape the model would never generate.

Minimum-variance hedge

$$h_t^* = \arg \min_h \text{Var}_t(\Delta V_{t,t+1} - h^\top \Delta F_{t,t+1}) = \Sigma_{FF,t}^{-1} \Sigma_{FV,t} \quad (24)$$

$$\Delta \Pi_{t,t+1} = \Delta V_{t,t+1} - h_t^\top \Delta F_{t,t+1} - TC(h_t - h_{t-1}) \quad (25)$$

The residual distribution should be decomposed into basis, constraint, model, execution, and policy error. A small delta is not evidence of low risk if local basis or availability dominates the residual.

Planned risk views

- 01 Monthly delta ladder with prompt, winter, and summer buckets
- 02 Hedged versus unhedged P&L distributions and expected shortfall
- 03 Inventory fan chart, action frequencies, and ratchet utilization
- 04 Value sensitivity surface across spread vol, mean reversion, and basis vol

11 • MODEL BOUNDARY

What exists now, and what changes for valuation

The runnable package is a research scaffold. The final valuation model will preserve its state and policy interfaces while replacing the main simplifying assumptions.

Time grid

24 monthly steps

Daily nomination calendar and no-flow days

Curve

13 simulated tenors; prompt OU + spread factor

Arbitrage-consistent multifactor forward dynamics calibrated to market

Volatility

Latent volatility proxy

Option-implied level, term structure, and factor correlation

Control

Maximum inject / hold / maximum withdraw

Continuous or contract nomination increments

Ratchets

Smooth inventory-dependent rates

Contract tables, outages, fuel, and losses

Basis

Mean-reverting local additive factor

Location-specific history, jumps, and hedge mapping

Validation

LSMC, NumPy DQN, heuristic comparison

Cross-fit LSMC, RL challenger, dual bounds, and hedge backtest

12 • NEXT BUILD

Plan for the final valuation and insight layer

Each phase has a numerical acceptance gate. The website will only promote a result to an insight after the policy, valuation, and hedge diagnostics agree.

PHASE 1

Calibrate the market and facility

Load the initial curve, estimate factor covariance and mean reversion, specify local basis, and encode contract ratchets and costs.

Gate: simulated moments, spread distributions, and ratchet tables reproduce calibration targets.

PHASE 2

Establish the value benchmark

Compute rolling intrinsic, cross-fit LSMC, and a frozen-policy out-of-sample value with standard errors and terminal-value sensitivity.

Gate: stable estimates across seeds, folds, basis families, and inventory grids.

PHASE 3

Train the RL challenger

Use the same paths, rewards, action constraints, and terminal rule. Compare policy surfaces and lower-bound values rather than training rewards.

Gate: RL outperforms simple policies and remains economically coherent under stress.

PHASE 4

Build hedges and risk attribution

Produce forward deltas, minimum-variance hedge ladders, transaction-cost P&L, basis residuals, and tail-risk scenarios.

Gate: hedge results hold on unseen paths and remain robust to factor and basis misspecification.

PHASE 5

Publish evidence-backed findings

Add interactive value decomposition, sensitivity surfaces, policy maps, inventory fans, and hedged P&L distributions to this site.

Gate: every chart states its measure, scenario, units, confidence interval, and model version.

SOURCES

Primary references

The specification follows the classical LSMC storage literature and connects it to modern statistical learning, deep optimization, and hedging methods.

⁰¹ Longstaff, F. A. & Schwartz, E. S. (2001). **Valuing American Options by Simulation: A Simple Least-Squares Approach** [↗](#). *Review of Financial Studies*, 14(1), 113-147.

⁰² Boogert, A. & de Jong, C. (2008). **Gas Storage Valuation Using a Monte Carlo Method** [↗](#). *Journal of Derivatives*, 15(3), 81-98.

03

Boogert, A. & de Jong, C. (2011). **Gas Storage Valuation Using a Multifactor Price Process** ↗. *Journal of Energy Markets*, 4(4).

04

Warin, X. (2012). **Gas Storage Hedging** ↗. In *Numerical Methods in Finance*, 421-445.

05

Ludkovski, M. & Maheshwari, A. (2020). **Simulation Methods for Stochastic Storage Problems: A Statistical Learning Perspective** ↗. *Energy Systems*, 11, 377-415.

06

Curin, N. et al. (2021). **A Deep Learning Model for Gas Storage Optimization** ↗. *Decisions in Economics and Finance*, 44, 1021-1037.

07

Buehler, H., Gonon, L., Teichmann, J. & Wood, B. (2019). **Deep Hedging** ↗. *Quantitative Finance*, 19(8), 1271-1291.
